

Formula sheet for Quiz 7

Note to students: you will be handed a copy with the quiz, so nothing here needs memorizing. It lists more than any one question needs, so deciding which applies is still your job.

Boxes

A box is (x_1, y_1, x_2, y_2) in **pixels**, top-left and bottom-right, which is what **torchvision** detectors return. Its area is $(x_2 - x_1)(y_2 - y_1)$.

Intersection over Union

For boxes A and B , the intersection box runs from $(\max(x_1^A, x_1^B), \max(y_1^A, y_1^B))$ to $(\min(x_2^A, x_2^B), \min(y_2^A, y_2^B))$:

$$I = \max(x_2 - x_1, 0) \cdot \max(y_2 - y_1, 0) \quad \text{IoU} = \frac{I}{\text{area}(A) + \text{area}(B) - I}$$

The union subtracts I because adding the two areas counts the overlap twice. IoU is 0 for boxes that do not touch and 1 for identical boxes. The same formula works on masks with I the count of pixels in both.

Anchor boxes

An anchor is a box of fixed size and position; the network predicts four **offsets** from it rather than coordinates. For an anchor (x_a, y_a, w_a, h_a) given as center, width and height, and a target box (x, y, w, h) :

$$t_x = \frac{x - x_a}{w_a}, \quad t_y = \frac{y - y_a}{h_a}, \quad t_w = \log \frac{w}{w_a}, \quad t_h = \log \frac{h}{h_a}$$

Dividing by the anchor size makes the targets scale-free; the log keeps the predicted width and height positive. The log is natural, inverted by exp.

Segmentation metrics

Over the pixels of one image, with TP, FP and FN counted for one class:

$$\text{pixel accuracy} = \frac{\text{pixels correct}}{\text{pixels total}} \quad \text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \quad \text{Dice} = \frac{2 \text{TP}}{2 \text{TP} + \text{FP} + \text{FN}}$$

Dice as a differentiable loss on predicted probabilities p and targets y , and the combination that is usually used:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum_i p_i y_i + \epsilon}{\sum_i p_i + \sum_i y_i + \epsilon} \quad \mathcal{L} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{Dice}}$$

Convolution output size

For one spatial axis, with input size W , kernel k , padding p and stride s , forward and then for a transposed convolution:

$$W_{\text{out}} = \left\lfloor \frac{W + 2p - k}{s} \right\rfloor + 1 \quad \text{ConvTranspose2d: } W_{\text{out}} = s(W - 1) - 2p + k$$

ConvTranspose2d(k=2, s=2) and **(k=4, s=2, p=1)** both exactly double the size. Parameter counts: **Conv2d**($C_{\text{in}}, C_{\text{out}}, k$) = $C_{\text{out}}(C_{\text{in}}k^2 + 1)$, **Linear**($n_{\text{in}}, n_{\text{out}}$) = $n_{\text{out}}(n_{\text{in}} + 1)$.

Notation

N batch size, C channels or classes, H and W spatial size, k kernel size, p padding, s stride, η the learning rate, ϵ a small constant that keeps a denominator away from zero.